Soko.market identity execution

Section X: Trusted Identity Execution Layer (TIEL)

Purpose

The Trusted Identity Execution Layer (TIEL) extends the Soko.market runtime with a hardware-backed identity architecture inspired by cellular networks. Rather than treating users, merchants, suppliers, customers, devices, and AI agents as application accounts, TIEL treats every participant as a network identity.

Unlike traditional authentication systems that rely on usernames, passwords, or bearer tokens, TIEL establishes cryptographically verifiable identities rooted in trusted hardware where available.

This architecture provides the foundation for a decentralized commerce network in which identities are provisioned, authenticated, updated, and revoked similarly to subscriber identities in telecommunications systems.

Design Philosophy

Traditional platforms ask:

"Who is logging in?"

TIEL asks:

"Which trusted identity is requesting access?"

Identity is therefore independent of:

- passwords
- devices
- sessions
- applications

Instead, every entity possesses a persistent cryptographic identity.

Core Components

1. Identity Authority (IA)

Equivalent to the Home Subscriber Server (HSS) in a cellular network.

Responsibilities:

- Issue identities
- Generate cryptographic credentials
- Maintain identity registry
- Revoke compromised identities
- Rotate credentials
- Manage trust relationships

Every issued identity receives:

- Global Identity ID (GIID)
- Public Key
- Private Key
- Identity Certificate
- Metadata
- Permissions

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2. Trusted Identity Module (TIM)

The TIM is the software equivalent of an eSIM profile.

Responsibilities:

- Securely stores private keys
- Performs cryptographic operations
- Never exposes secret keys
- Maintains active identity profile
- Supports multiple identities

Whenever hardware permits, TIM executes inside:

- ARM TrustZone

- Apple Secure Enclave
- Android Hardware Keystore
- TPM
- Secure Element

Otherwise it falls back to encrypted software storage.

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3. Trusted Execution Environment

Whenever supported by the device, cryptographic operations occur inside the Trusted Execution Environment (TEE).

Functions:

- Key generation
- Signature generation
- Encryption
- Challenge-response authentication
- Secure storage
- Remote attestation

Private keys never leave the secure environment.

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4. Identity Profiles

Each device may contain multiple identities.

Example:

Merchant Identity

Supplier Identity

Customer Identity

Warehouse Agent

Delivery Agent

Inventory AI

Accounting AI

Marketplace Administrator

Profiles may be activated without recreating identities.

Identity Lifecycle

Provision

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Generate Keys

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Issue Certificate

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Install Identity

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Activate

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Authenticate

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Operate

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Rotate

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Suspend

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Revoke

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Archive

Authentication Flow

Instead of passwords:

1. Device requests authentication.
2. Identity Authority sends random challenge.
3. TIM signs challenge.
4. Authority verifies signature.
5. Session keys established.
6. Secure encrypted session begins.

No password is transmitted.

No private key is transmitted.

Remote Attestation

Where supported, devices prove:

- genuine hardware
- approved firmware
- approved Sokoclaw runtime
- uncompromised execution environment

Attestation allows Soko.market to trust both the identity and the integrity of the runtime.

Identity Types

Human Identity

Merchant Identity

Supplier Identity

Customer Identity

AI Agent Identity

Business Identity

Organization Identity

Store Identity

Warehouse Identity

Vehicle Identity

IoT Device Identity

Payment Terminal Identity

Each identity follows the same lifecycle and cryptographic standards.

Device Trust Levels

Level 0

Software-only storage.

Level 1

Hardware-backed Keystore.

Level 2

Trusted Execution Environment.

Level 3

Dedicated Secure Element.

Level 4

Certified Trusted Identity Module.

Sensitive operations may require higher trust levels.

Identity Roaming

Identities are portable.

Users may securely move identities between devices without changing their logical identity.

Lost devices can be remotely revoked.

New devices can receive securely provisioned identities.

AI Agent Identities

Every Sokoclaw agent possesses its own identity.

The identity signs:

- generated workflows
- transactions
- tool invocations
- memory updates
- purchased skills
- marketplace packages

This enables accountability and traceability across the ecosystem.

Runtime Integration

The Trusted Identity Execution Layer becomes the first stage of the Sokoclaw runtime.

Boot

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Hardware Verification

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Trusted Identity Module

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Identity Authentication

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Remote Attestation

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Permission Loading

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Model Router

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Skill Engine

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Planner

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Execution Engine

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Memory System

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Synchronization

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Audit Logging

Every runtime execution therefore begins from a verified identity rather than from an authenticated application session.

Long-Term Vision

TIEL transforms Soko.market from an application platform into an identity-centric commerce network.

In this model:

- every participant is a cryptographically verifiable network identity;
- every AI agent is a trusted computational identity;
- every transaction is cryptographically attributable;
- trust is established through hardware-backed identity rather than passwords;
- the platform operates with principles inspired by telecommunications identity systems while remaining independent of cellular infrastructure.

This architecture forms the foundation for secure multi-tenant commerce, trusted AI execution, offline-first operation, and future interoperability between independent Soko.market deployments.